Year 4 Mathematics Solutions

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



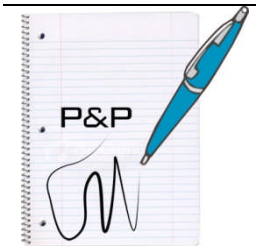
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



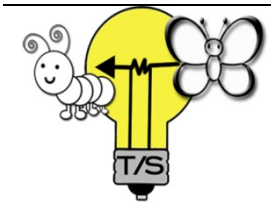
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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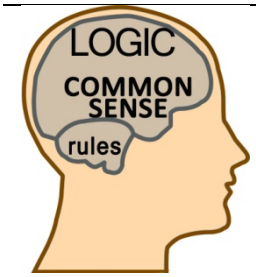
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

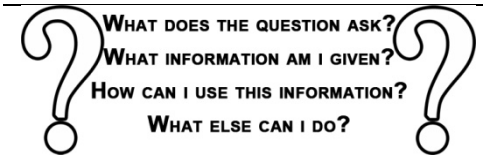
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 4 Mathematics

Number

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Exercise 1

Representing Numbers

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1) Write as numbers

a) Three hundred and ninety

390

b) Eight hundred and eighty three

883

c) Seven hundred and ninety three

793

d) Five hundred and six

506

e) Nine hundred and nine

909

2) Write as numbers

a) Two thousand two hundred and three

2203

b) Seven thousand four hundred and ninety seven

7497

c) Eight thousand six hundred and thirty

8630

d) Nine thousand and twenty one

9021

e) Three thousand and one

3001

3) Write in words

a) 2713

Two thousand seven hundred and thirteen

b) 2097

Two thousand and ninety seven

c) 3330

Three thousand three hundred and thirty

d) 8090

Eight thousand and ninety

e) 2010

Two thousand and ten

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f) 1117

One thousand one hundred and seventeen

g) 0

Zero

4) Write down the number that comes before each of these numbers

a) 331

330

b) 156

155

c) 905

904

d) 120

119

e) 1710

1709

f) 1100

1099

g) 2442

2441

h) 1900

1899

i) 9001

9000

j) 3006

3005

k) 1234

1233

l) 10000

9999

5) Write the number that comes after each of these numbers

a) 819

820

b) 1090

1091

c) 8881

8882

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d) 4223

4224

e) 8010

8011

f) 711

712

g) 1999

2000

h) 3009

3010

6) Put these numbers in order from smallest to largest

1325, 1101, 1123, 3000, 2946, 2121, 1015, 2221, 2323, 9104, 694

694, 1015, 1101, 1123, 1325, 2121, 2221, 2323, 2946, 3000, 9104

7) Put these numbers in order from largest to smallest.

2015, 2004, 4020, 1912, 1911, 2333, 3322, 2921, 2221, 4121, 3004

4121, 4020, 3322, 3004, 2921, 2333, 2221, 2015, 2004, 1912, 1911

8) What is the value of the number 4 in each of these numbers?

a) 1034

Ones

b) 1435

Hundreds

c) 2114

Units

d) 4027

Thousands

e) 4

Units

f) 1040

Tens

g) 2047

Tens

9) Use the > or < sign to show the relationship between the following pairs of numbers

a) 1234 < 2134

b) 9821 > 9281

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c) 8005 < 8015

d) 1023 > 103

e) 970 > 907

f) 1099 > 1089

10) Write the number that is 10 less than the number shown. Repeat 4 times

a) 675

665, 655, 645, 635, 625

b) 555

545, 535, 525, 515, 505

c) 390,

380, 370, 360, 350, 340

d) 442

432, 422, 412, 402, 392

e) 530

520, 510, 500, 490, 480

f) 401

391, 381, 371, 361, 351

g) 112

102, 92, 82, 72, 62

h) 220

210, 200, 190, 180, 170

i) 1039

1029, 1019, 1009, 999, 989

j) 1050

1040, 1030, 1020, 1010, 1000

k) 908

898, 888, 878, 868, 858

11) Write the number that is 10 more than the number shown. Repeat four times

a) 1121

1131, 1141, 1151, 1161, 1171

b) 2020

2030, 2040, 2050, 2060, 2070

c) 3175

3185, 3195, 3205, 3215, 3223

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d) 1099

1109, 1119, 1129, 1139, 1149

e) 803

813, 823, 833, 843, 853

f) 960

970, 980, 990, 1000, 1010

g) 999

1009, 1019, 1029, 1039, 1049

h) 100

110, 120, 130, 140, 150

i) 1251

1261, 1271, 1281, 1291, 1301

12) Round the following numbers to the nearest thousand, hundred and ten

a) 1263

1000, 1300, 1260

b) 926

1000, 900, 930

c) 101

0, 100, 100

d) 4565

5000, 4600, 4560

e) 8555

9000, 8600, 8560

f) 7550

8000, 7600, 7550

g) 6005

6000, 6000, 6000

h) 1111

1000, 1100, 1110

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Exercise 2

Addition & Subtraction

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1) Add these numbers

a) 632 + 114

746

b) 247 + 319

566

c) 621 + 535

1156

d) 877 + 223

1100

e) 135 + 175

310

f) 414 + 441

855

2) Add these numbers

a) 2225 + 529

2754

b) 4302 + 410

4712

c) 8009 + 377

8386

d) 4335 + 323

4658

e) 8122 + 110

8232

f) 9334 + 73

9407

3) Subtract these numbers

a) 816 – 412

404

b) 594 – 482

112

c) 756 -511

245

d) 929 – 353

576

e) 504 – 127

377

f) 865 – 821

44

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g) 9026 – 312

8714

h) 6111 – 3227

2884

4) Peter has 840 stamps, John has 275 stamps. How many stamps do they have between them?

$840 + 275 = 1115$ stamps

5) Alan weighs 145 kg, Chris weighs 148 kg. How much do they weigh together?

$145 + 148 = 293$ kg

6) There were 1510 more people at the football game than at the rugby. If there were 4600 people at the football how many people were at the rugby?

$4600 - 1510 = 3090$ people at the rugby

7) Tom and Jerry have read 410 books between them. If Tom has read 318 books, how many books has Jerry read?

$410 - 318 = 92$ books

8) 138 students passed a test, 112 failed, and 35 were absent. How many students are in the school?

$138 + 112 + 35 = 285$ students

9) What number is 299 less than 6075?

$6075 - 299 = 5776$

10) What is the difference between 2710 and 3244?

$3244 - 2710 = 534$

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Exercise 3

Multiplication & Division

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1) Calculate the following

a) 5×10

50

b) 5×20

100

c) 5×30

150

d) 40×5

200

e) 60×5

300

f) 20×7

140

g) 40×7

280

h) 60×7

420

i) 60×9

540

2) Calculate the following

a) 8×13

104

b) 16×9

144

c) 11×7

77

d) 17×8

126

e) 32×6

192

f) 45×9

405

3) Calculate the following

a) 15×6

90

b) 15×8

120

c) 6×15

90

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d) 7×15

105

e) 9×15

135

f) From your answers, state a method for quickly multiplying any number by 15

The answer is ten times the number plus half of the result

4)

a) How many fours in 24?

6

b) What is 24×25?

600

c) How many fours in 28?

7

d) What is 28×25?

700

e) How many fours in 32?

8

f) What is 32×25?

800

g) Use your answers to parts a to f to state a method for quickly multiplying any number by 25

The answer is the amount of fours in the number times one hundred

5) Calculate the following

a) $24 \div 5$

$4\frac{4}{5}$

b) $33 \div 8$

$4\frac{1}{8}$

c) $15 \div 4$

$3\frac{3}{4}$

d) $35 \div 7$

5

e) $24 \div 7$

$3\frac{3}{7}$

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f) $74 \div 7$

$10\frac{4}{7}$

g) $37 \div 5$

$7\frac{2}{5}$

h) $49 \div 8$

$6\frac{1}{8}$

i) $21 \div 4$

$5\frac{1}{4}$

j) $82 \div 8$

$10\frac{2}{8} = 10\frac{1}{4}$

6) Write the factors of the following

a) 9

1, 3, 9

b) 15

1, 3, 5, 15

c) 24

1, 2, 3, 4, 6, 8, 12, 24

d) 7

1, 7

e) 4

1, 2, 4

f) 1

1

g) 64

1, 2, 4, 8, 16, 32, 64

h) 100

1, 2, 4, 5, 10, 20, 25, 50, 100

i) 22

1, 2, 11, 22

7) Mary has 40 lollies. If she gives each of her 6 friends an equal amount of lollies, how many will she have left over for herself? (She gives each friend the most that she can)

The number closest to 40 that is a multiple of 6 is 36; this leaves 4 lollies for Mary

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8) Alan buys 5 pens and gets 5 cents change from his dollar. How much was each pen?

$1 – 5 cents = 95 cents. Each pen was $\frac{95}{5} = 19$ cents

9) Kathy is having a birthday party and wants each friend to get five lollies in their party bag. If there are 8 friends coming to the party, how many lollies will be left over from a bag of 50?

Each friend gets 5 lollies x 8 friends = 40 lollies. This leaves 10 lollies.

10) Tom has $5 left after giving an equal amount of money to a number of charities. If he started with $35, list how many charities he may have given money to, and how much he would have given to each.

He gave $35 - 5 = \$30$. He could have given any combination that makes $30

1 charity x $30
2 charities x $15
3 charities x $10
4 charities x $7.50
5 charities x $6
6 charities x $5
8 charities x $3.75
10 charities x $3
12 charities x $2.50
15 charities x $2
20 charities x $1.50
24 charities x $1.25

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Exercise 4

Number Patterns

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1) Find the sixth term in the following sequences

a) 3, 6, 9, 12

Add 3 each time, so 5th term is 15, 6th term is 18

b) 2, 4, 6

Add 2 each time, so 4th term is 8, 5th term is 10, 6th term is 12

c) 5, 10, 15

Add 5 each time, so 4th term is 20, 5th term is 25, 6th term is 30

d) 7, 14, 21

Add 7 each time, so 4th term is 28, 5th term is 35, 6th term is 42

e) 4, 8, 12

Add 4 each time, so 4th term is 16, 5th term is 20, 6th term is 24

f) 9, 18, 27

Add 9 each time, so 4th term is 36, 5th term is 45, 6th term is 54

2) Find the fifth term in the following sequences

a) 25, 20, 15

Subtract 5 each time, so 4th term is 10, 5th term is 5

b) 40, 32, 24

Subtract 8 each time, so 4th term is 16, 5th term is 8

c) 63, 54, 45

Subtract 9 each time, so 4th term is 36, 5th term is 27

d) 63, 60, 57

Subtract 3 each time, so 4th term is 54, 5th term is 51

e) 14, 11, 8, ___, ___

Subtract 3 each time, so 4th term is 5, 5th term is 2

3) Find the missing numbers

a) $\square + 12 = 20$

8

b) $\square + 10 = 20$

10

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c) $\square \times 5 = 30$

6

d) $11 \times \square = 44$

4

e) $7 + \square = 15$

8

f) $\square \times 3 = 21$

7

g) $\square + 10 = 15$

5

4) Complete the following sequences

a) $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$, ____, ____

Add $\frac{1}{4}$ each time, so next two terms are $\frac{5}{4}, \frac{7}{4}$

b) $\frac{1}{3}, \frac{2}{3}, 1$, ____, ____

Add $\frac{1}{3}$ each time so next two terms are $\frac{4}{3}, \frac{5}{3}$

c) $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}$, ____, ____

Add $\frac{1}{5}$ each time, so next two terms are $\frac{4}{5}, \frac{5}{5}$

d) $\frac{5}{3}, \frac{4}{3}, 1$, ____,____

$1 = \frac{3}{3}$, so subtract $\frac{1}{3}$ each time, so next two terms are $\frac{2}{3}, \frac{1}{3}$

e) $\frac{10}{7}, \frac{9}{7}, \frac{8}{7}$, _____,_____

Subtract $\frac{1}{7}$ each time, so next two terms are $\frac{7}{7}, \frac{6}{7}$

f) $\frac{97}{100}, \frac{98}{100}, \frac{99}{100}$, _____,_____

Add $\frac{1}{100}$ each time, so next two terms are $\frac{100}{100}, \frac{101}{100}$

5) Peter wants to give 8 people \$5 each. If he has \$32 how much more money does he need to be able to do this?

$8 \times \$5 = \$40$ so he needs an extra \$8

6) There are 9 tables in a restaurant. Each table has 6 chairs around them. If there are 70 people coming to the restaurant at one time, how many more chairs are needed?

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$9x6 = 54$, so will need another 16 chairs

7) Every minute 5 ants crawl out of an ant hill.

a) How many ants have crawled out after 4 minutes?

$4 \times 5 = 20$ ants.

b) There are 50 ants out of the ant hill. How many more minutes will go by until there are 75 ants out of the ant hill?

25 more ants will crawl out in 5 minutes

8) After 4 hours there were 24 cars in a car park. If the same number of cars park each hour

a) How many cars will be in the car park after 7 hours?

$24 \div 4 = 6$ so 6 cars park each hour. $7x6 = 42$ cars

b) How many hours will have passed until there are 54 cars in the car park?

$54 \div 6 = 9$ hours

c) If the car park holds 96 cars, how long until it is full from when it first opened?

$96 \div 6 = 16$ hours

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Exercise 5

Fractions

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1) Write the following as a fraction

a) One fifth

$\frac{1}{5}$

b) One tenth

$\frac{1}{10}$

c) Two fifths

$\frac{2}{5}$

d) One hundredth

$\frac{1}{100}$

e) Three fifths

$\frac{3}{5}$

f) Three tenths

$\frac{3}{10}$

g) Seventeen hundredths

$\frac{17}{100}$

h) Four fifths

$\frac{4}{5}$

i) Nine tenths

$\frac{9}{10}$

2) Write the following in words

a) $\frac{1}{5}$

One fifth

b) $\frac{1}{100}$

One hundredth

c) $\frac{3}{10}$

Three tenths

d) $\frac{11}{100}$

Eleven hundredths

e) $\frac{7}{10}$

Seven tenths

f) $\frac{4}{5}$

Four fifths

g) $\frac{99}{100}$

Ninety nine hundredths

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3) Put these fractions in order from smallest to largest

$\frac{3}{5}, \frac{2}{5}, \frac{4}{5}, \frac{1}{5}$

$\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$

4) Put these fractions in order from largest to smallest

$\frac{5}{10}, \frac{1}{10}, \frac{7}{10}, \frac{2}{10}, \frac{6}{10}$

$\frac{7}{10}, \frac{6}{10}, \frac{5}{10}, \frac{2}{10}, \frac{1}{10}$

5) Fill in the missing numbers

$\frac{97}{100}, \frac{95}{100}, \frac{93}{100}, \frac{91}{100}, __, __$

Each fraction reduces by $\frac{2}{100}$, so

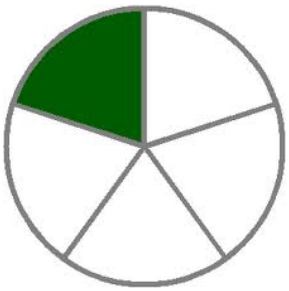
next two terms are $\frac{89}{100}, \frac{87}{100}$

6) Fill in the missing numbers

$\frac{11}{5}, \frac{14}{5}, __, \frac{20}{5}, __, __$

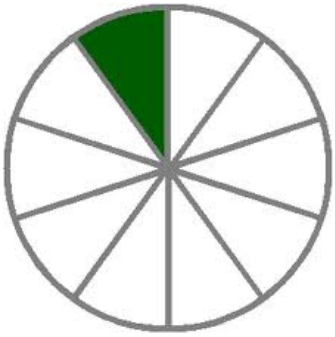
7) What fraction is shaded in the following diagrams?

a)



One part out of five $= \frac{1}{5}$

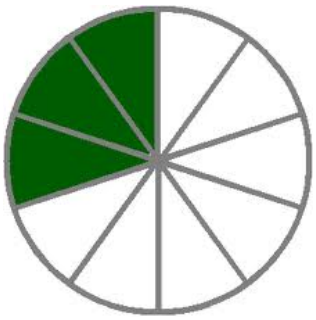
b)



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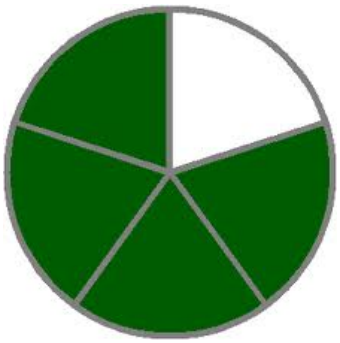
One part out of ten$= \frac{1}{10}$

c)



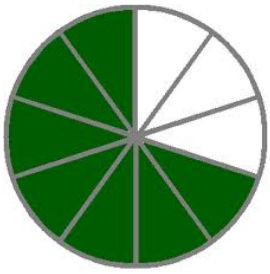
Three parts out of ten $= \frac{3}{10}$

d)



Four parts out of five $= \frac{4}{5}$

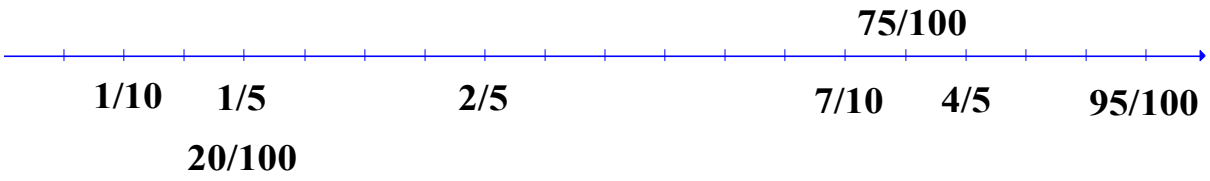
e)



Seven parts out of ten $= \frac{7}{10}$

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8) Place the fractions $\frac{1}{10}, \frac{1}{5}, \frac{20}{100}, \frac{2}{5}, \frac{7}{10}, \frac{75}{100}, \frac{4}{5}, \frac{95}{100}$ on a number line



9) Tim has one fifth of his lollies left, while Jack has eaten two fifths. Who has more lollies left?

If Jack has eaten $\frac{2}{5}$, then he has $1 - \frac{2}{5} = \frac{3}{5}$ of his lollies left, which is more than $\frac{1}{5}$

10) Peter had \$100 and spent \$50. Jack had \$10 and spent only \$3. Who spent the bigger fraction of their money?

Peter spent $\frac{1}{2}$ of his money, Jack spent $\frac{3}{10}$. On a number line $\frac{1}{2} > \frac{3}{10}$ so Peter spent the bigger fraction

11) A fly spray kills two fifths of the flies in a room, whilst another kills three tenths of them. Which fly spray works better?

On a number line $\frac{2}{5} > \frac{3}{10}$ so the first fly spray works better

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Exercise 6

Decimals & Percentages

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1) Round the following decimals to the nearest whole number

a) 1.48

1

b) 11.05

11

c) 13.74

14

d) 0.22

0

e) 1.55

2

f) 22.51

23

2) Express the following fractions and mixed numbers as decimals

a) $\frac{3}{10}$

0.3

b) $\frac{15}{100}$

0.15

c) $3\frac{1}{10}$

3.1

d) $1\frac{7}{10}$

1.7

e) $1\frac{7}{100}$

1.07

f) $1\frac{77}{100}$

1.77

3) Multiply each of the following by 10

a) 1.4

14

b) 2.5

25

c) 3.7

37

d) 5.8

58

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e) 10.2

102

f) 1.36

13.6

g) 2.45

24.5

h) 6.22

62.2

i) 8.49

84.9

j) 15.43

154.3

4) Multiply each of the following by 100

a) 1.52

152

b) 2.75

275

c) 4.26

426

d) 8.04

804

e) 13.11

1311

f) 8.6

860

g) 7.2

720

h) 4.3

430

i) 1.2

120

5) Write the following as a decimal

a) 30%

0.3

b) 15%

0.15

c) 20%

0.2

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d) 10%

0.1

e) 75%

0.75

f) 90%

0.9

g) 100%

1.0

6) Write the following as a fraction

a) 50%

$\frac{1}{2}$

b) 25%

$\frac{1}{4}$

c) 10%

$\frac{1}{10}$

7) Divide each of the following by 10

a) 13.2

1.32

b) 10.8

1.08

c) 9.6

0.96

d) 7.2

0.72

e) 3.3

0.33

f) 1

0.1

8) Divide each of the following by 100

a) 152.5

1.525

b) 143.2

1.432

c) 131.9

1.319

d) 106.5

1.065

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e) 98.9

0.989

f) 90.2

0.902

g) 66.6

0.666

h) 9.25

0.925

9) Alex has $14.25 in his bank account. Tom has ten times as much. How much money does Tom have?

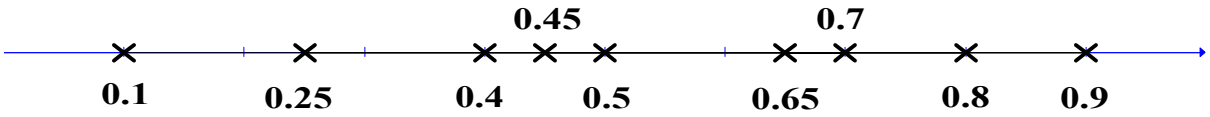
$\$14.25 \times 10 = \$142.50$

10) John runs 30km and Jill runs 50% of that distance. How far did Jill run?

$50\% \times 30km = 15km$

11) Place the following decimals on a number line

0.7, 0.65, 0.8, 0.1, 0.25, 0.4, 0.5, 0.9, 0.45



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12) Express the following as a decimal

a) $\frac{51}{1000}$

0.051

b) $\frac{74}{1000}$

0.074

c) $\frac{17}{1000}$

0.017

d) $\frac{7}{1000}$

0.007

e) $\frac{1}{1000}$

0.001

13) Calculate the following

a) 1.2 + 3.4

4.6

b) 3.6 + 4.3

7.9

c) 10.2 + 5.3

15.5

d) 1.25 + 3.1

4.35

e) 2.56 + 5.2

7.76

f) 7.4 + 2.22

9.62

g) 8.1 + 3.05

11.15

14) Calculate the following

a) $7.4 - 2.3$

5.1

b) $9.6 - 3.1$

6.5

c) $10.7 - 9.6$

1.1

d) $8.4 - 4.8$

3.6

e) $3.2 - 2.5$

0.7

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f) $7.65 - 4.3$

3.35

g) $3.43 - 2.3$

1.13

h) $5.69 - 3.06$

2.63

i) $7.32 - 5.61$

1.71

j) $8.19 - 5.43$

2.76

15) Jake has \$14.70 and spends \$12.35. How much money does he have left?

$14.7 - 12.25 = \$2.35$

16) Paul has \$12.35 and his grandfather gives him \$11.15. How much money does Paul now have?

$12.35 + 11.15 = \$23.50$

17) Barbara wants to save up to buy a new dress that costs \$35.30. At the moment she has \$16.10. How much more money does she need to be able to buy the dress?

$35.30 - 16.10 = \$19.20$

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Exercise 7

Chance

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1) Alan tosses two coins. List the possible combinations they could land on

Both coins heads

First coin heads, second coin tails

First coin tails, second coin heads

Both coins tails

2) Peter rolls two dice and adds the two numbers. List all the numbers that he could get

2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

3) List what the two dice from question 2 could show to get a total of 7

First dice 1 + second dice 6

First dice 2 + second dice 5

First dice 3 + second dice 4

First dice 4 + second dice 3

First dice 5 + second dice 2

First dice 6 + second dice 1

4) List what the two dice from question 2 could show to get a total of 12

First dice 6 + second dice 6

5) There are 6 red shirts, 6 blue shirts and 6 yellow shirts in a draw. If a boy pulls a shirt out without looking:

a) List what colour shirt he might pull out

Red, blue or yellow

b) Which colour shirt will he probably pull out?

Could pull any colour

c) Could he pull out 6 yellow shirts in a row?

Yes, there are 6 yellow shirts so he could pull all of them out in a row

6) There are 20 red, 20 blue and 20 green lollies in a jar. If Jack closes his eyes and chooses one:

a) What colour lolly will he probably choose?

Could choose red, blue or green

b) What colour lolly could he not get?

Any colour but the above

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c) If he pulls out a red lolly first time, will he definitely get a red lolly next time?

No: he could get a red lolly, but not definitely

d) Could he pull out 20 red lollies in a row?

Yes: there are 20 red lollies in the jar so he could pull them all out in a row

e) If he did this, which colour would he be more likely to pull out in his next turn?

Could then pull out blue or green

7) In a jar there are 20 blue buttons. In another jar there are 20 blue and 20 yellow buttons.

a) Which jar has more blue buttons?

Each jar has the same number of blue buttons

b) From which jar is he more likely to pull out a blue button?

The jar with only blue buttons

c) Is he more likely to pull a yellow or blue button from the second jar?

Either is equally likely

d) Could he pull 20 yellow buttons in a row from the second jar

Yes, there are 20 yellow lollies in the jar so he could pull 20 out in a row

e) If he did this, from which jar would he then have more chance of pulling a blue button from?

Both jars would have only 20 buttons so both would have equal chance

8) Of the following events, which are certain to happen, impossible, or could happen?

a) The sun will rise tomorrow

Certain

b) You will eat food

Certain

c) You will go to school

Could happen (if not holidays or a weekend etc)

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d) You will get every maths question right

Could happen

e) You will turn 45 years old tomorrow

Could happen

f) Everyone in your class will win a million dollars tomorrow

Impossible

g) You will ride a bicycle

Could happen

9) Tom rolls two normal 6 sided dice and adds the numbers. Which total is he most likely to get?

There are more ways to get a total of 7 than any other number

10) Alan tosses two coins; are they more likely to land on two heads or two tails?

Either combination is equally likely

11) Peter spins a spinner with 3 red and 3 white faces. If he spins it twice, list all the combinations of colours he could get

A red and a red

A red and a white

A white and a red

A white and a white

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Year 4 Mathematics

Data

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Exercise 1

Data Tables

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1) Tom made a table that shows how many of his classmates have each colour as their favourite

	Green	Yellow	Blue	White	Black
Girls	4	1	1	6	2
Boys	5	0	8	4	4

a) How many children in Tom's class?

Adding all the numbers gives 14 girls and 21 boys equals 35 in total

b) Which colour was most popular?

White had 10 votes

c) Which colour was most popular for boys?

Blue (8 votes)

d) Which colours had equal numbers of children voting for it?

Green and blue (9 votes)

e) Which colour or colours had equal number of boys voting for it?

White and black (4 votes)

2) A group of people was asked to vote for one day as their favourite day of the week

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Men	1	3	5	10	5	6	15
Women	3	0	2	5	11	3	15

a) How many people were asked?

Adding all the numbers gives 84

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b) What was most people's favourite day?

Sunday (30 votes)

c) Which day was the least favourite of women?

Tuesday (0 votes)

d) Which day had the biggest difference in the number of men and women voting for it?

Friday (5 men 11 women)

3) A man made a list of the cost of a type of blanket and a fan at different times of the year

	January	March	May	July	September	November
Blankets	$3.50	$4	$5	$6.50	$5	$4
Fans	$20	$18	$15	$10	$12	$14

a) In which of the months was the blanket the cheapest?

January ($3.50)

b) In which month was the fan dearest?

January ($20)

c) What was the difference in its price between a fan and a blanket in September?

$(\$12 - \$5 = \$7)$

d) In which month were the prices closest?

July $(\$10 - \$6.50 = \$3.50)$

e) Explain why the prices changed so much during the year?

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In summer people would buy more fans and fewer blankets, and in winter the opposite. This makes them dearer or cheaper

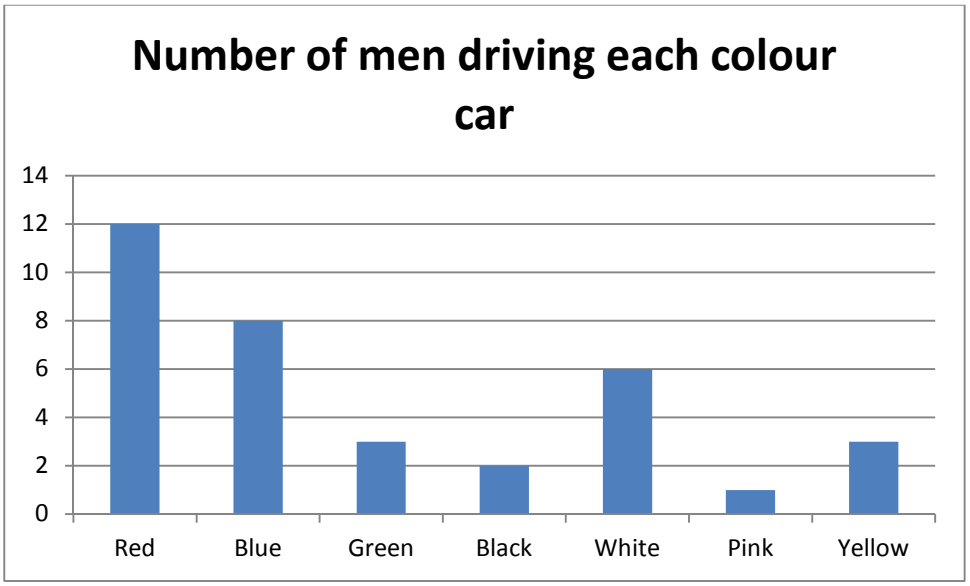
4) Show the following data in a two way table

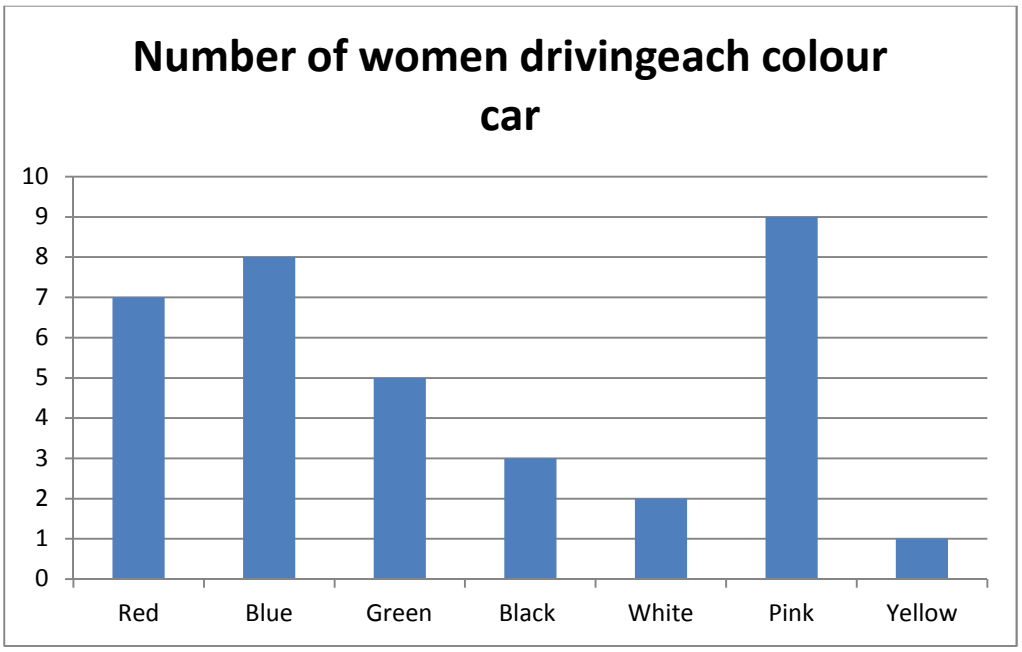
- 100 people were surveyed as to their favourite car
- Everyone had a choice of 4 cars
- 10 men said they like Holden best
- 15 women preferred Toyota
- 5 more men than women preferred Nissan
- 10 more women than men preferred Ford
- 20 men preferred Nissan
- 12 women preferred Ford
- Equal numbers of men and women were surveyed

	Holden	Toyota	Nissan	Ford
Men	10	18	20	2
Women	8	15	15	12

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5) The graphs show the number of people that own a certain colour car





a) Show the information in a two way table

	Red	Blue	Green	Black	White	Pink	Yellow
Men	12	8	3	2	6	1	3
Women	7	8	5	3	2	9	1

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b) How many people were surveyed?

70

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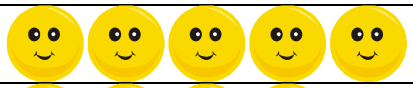
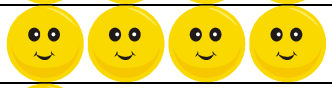
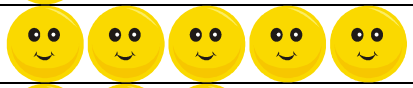
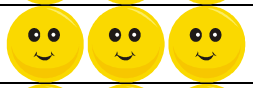
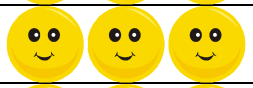
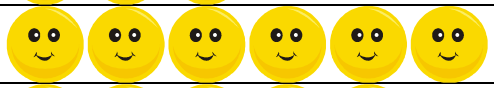
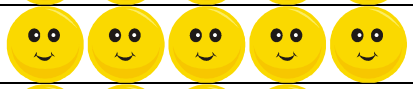
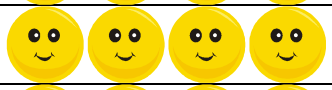
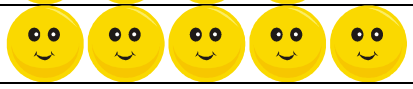
Exercise 2

Picture Graphs

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1) The picture graph below shows a sport and the number of children for whom it is their favourite

Each "face" represents 5 people

Game Number	Attendance
Football	
Rugby	
Soccer	
Basketball	
Hockey	
Swimming	
Tennis	
Golf	
Bowling	
Baseball	

a) Which sport is most popular?

Tennis

b) For how many people is it their favourite?

$6 \times 5 = 30$

c) For how many people is swimming their favourite sport?

$3 \times 5 = 15$

d) How many people were asked?

$41 \times 5 = 205$

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e) Is swimming or hockey more popular?

They are equally popular

2) Some people were asked how many times they ate fish. The picture graph shows their answers. Each fish represents 15 days of the year

Name	Number of days eating fish
Tom	
Benny	
Jane	
Julie	
Karen	
Brian	
Richard	
Ray	
Daniel	
Craig	

a) Who eats fish the most days of the year?

Jane

b) How many days a year do they eat fish?

$8 \times 15 = 120$

c) Who eats fish on the least number of days?

Richard

d) How many days do they eat fish on?

$2 \times 15 = 30$

e) If someone ate fish on 50 days of the year, how could you show this on the graph? Can you think of a better way to show numbers of days that are not groups of 15?

- Could make part of a fish equal to say 5 days

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- Could show a continuous bar instead of pieces
- Could use colours for different numbers

3) The graph below shows the number of kilos of each fruit bought in a week by a cafe. Bananas were \$2.50, apples \$2, oranges \$3, watermelon \$1.50 and strawberries \$4 per kilo



a) On which fruit did the cafe spend most money?

Strawberries (4kg x \$4 per kg = \$16)

b) What fruit did the cafe buy least of?

Oranges (2 kg)

c) How many kilos of fruit were bought in total?

17kg

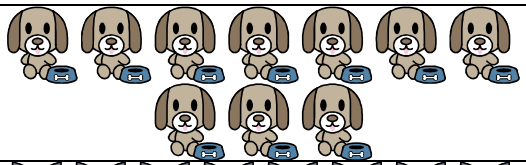
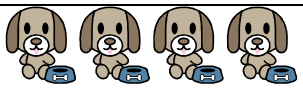
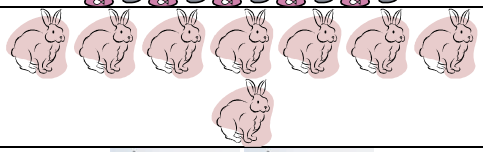
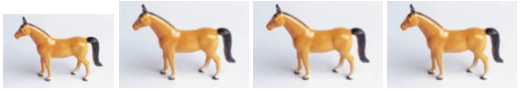
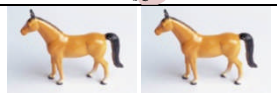
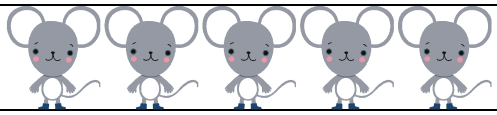
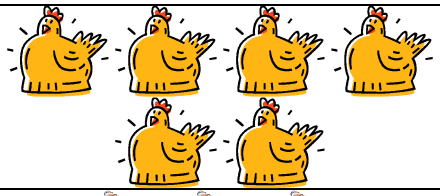
d) How much did the cafe spend on fruit in total?

$(5 \times \$2.50) + (3 \times \$2) + (2 \times \$3) + (3 \times \$1.50) + (4 \times \$4) = \$45$

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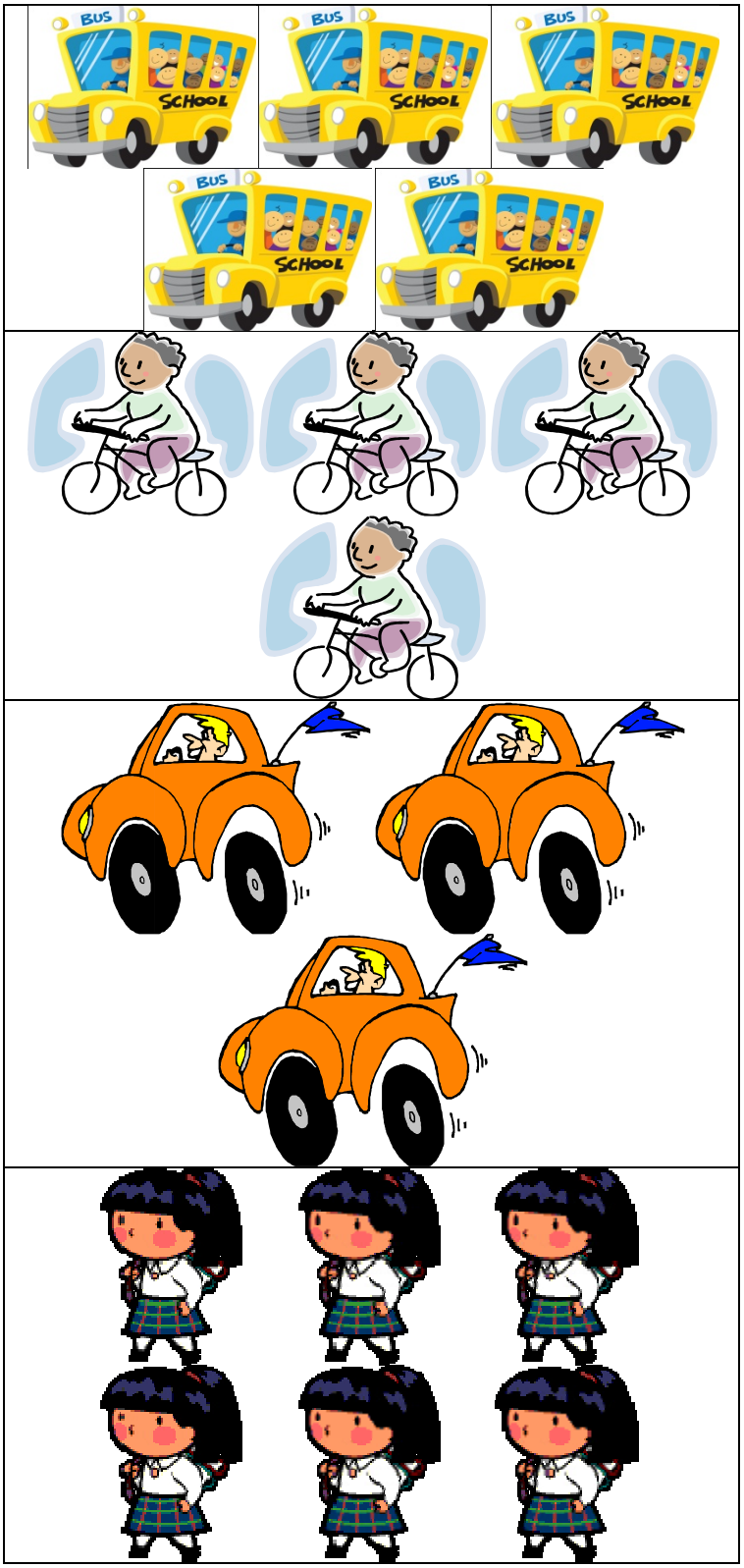
4) Draw a picture graph that shows the number of people that voted for their favourite animal

Animal	Number of men	Number of women
Dog	10	4
Cat	8	5
Rabbit	2	8
Horse	4	2
Mouse	5	0
Chicken	4	6
Lion	5	3
Tiger	3	1
Snake	1	0
Monkey	0	1

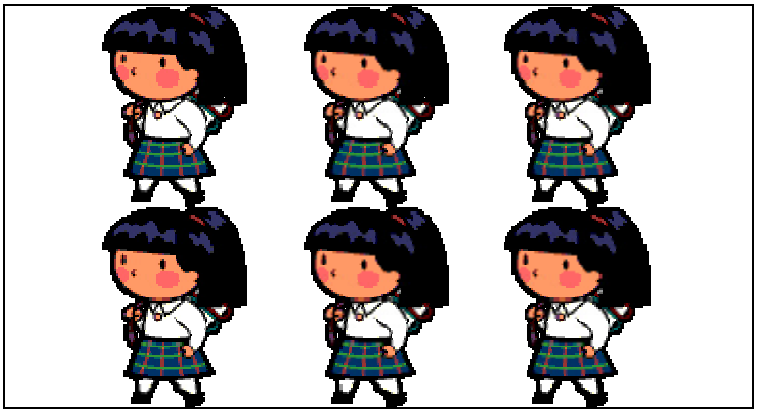
Number of men	Number of women
	
	
	
	
	
	
	
	
	
0	

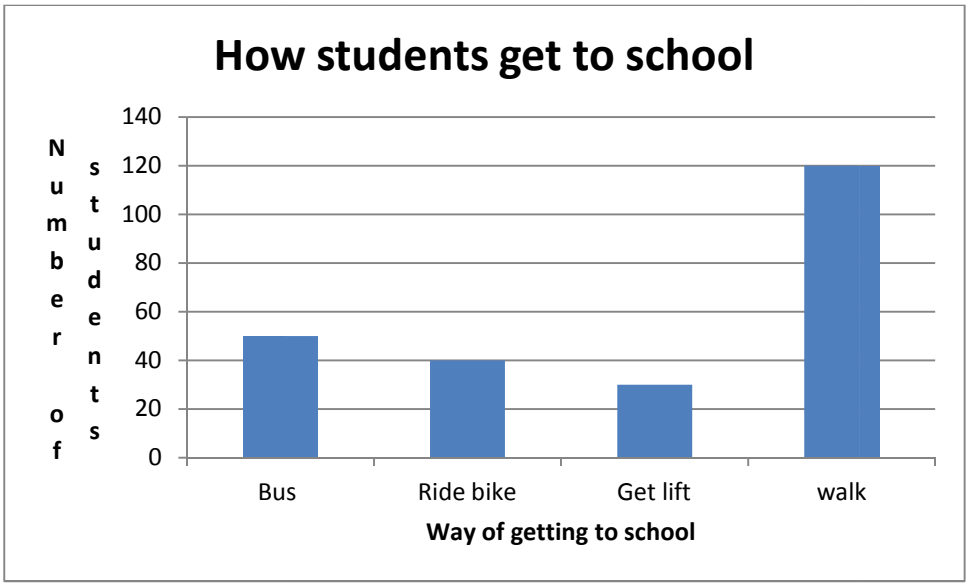
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5) The following picture graph shows the number of children that get to school in different ways. Each picture represents 10 children. Show the same information in a column graph



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Year 4 Mathematics

Space

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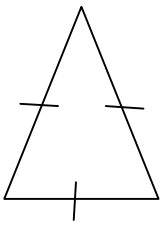
Exercise 1

Tessellations

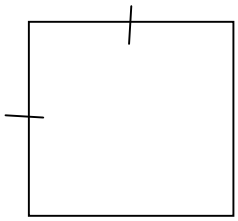
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1) Which of the following shapes tessellate?

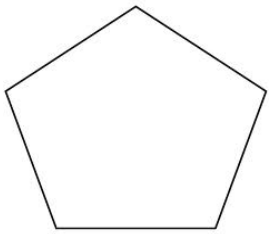
a)



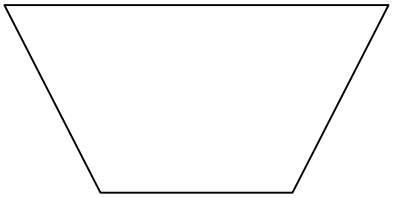
b)



c)



d)



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e)



All tessellate except shape c

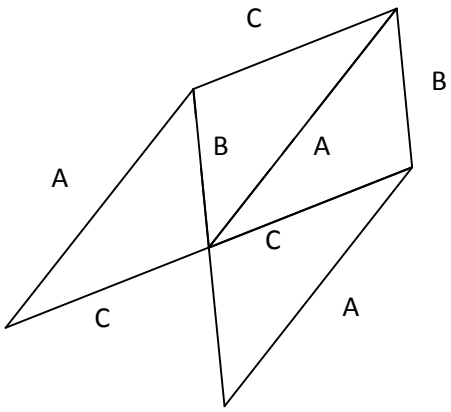
2) In the space in the table, write down how many of each shape is necessary to completely tessellate around a point

Equilateral Triangle	6
Square	4
Regular Pentagon	Cannot tessellate
Regular Hexagon	3

3) Explain in your own words why you need different numbers of certain shapes to be able to tessellate them

Because the angle inside each shape is a different size depending on which shape is chosen. So you need more or less of them to fill the same space

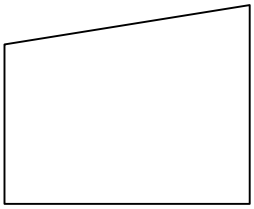
4) The side lengths of the triangle are all different. By rotating the triangle, construct a tessellation, and identify the side names in each triangle



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5) Using the triangle above, form a tessellation by using a combination of rotations and a reflection

6) By using rotations, construct a tessellation from the following quadrilateral

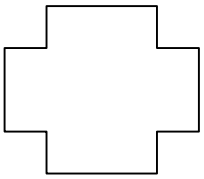


7) By using a translation (sliding), form a tessellation from the following shape

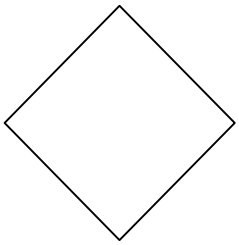


8) What technique(s) would you use to tessellate the following shapes?

a)

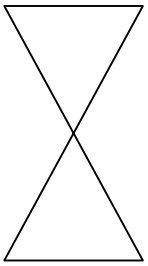


b)

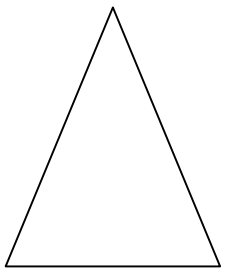


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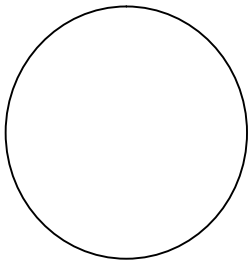
c)



d)



e)



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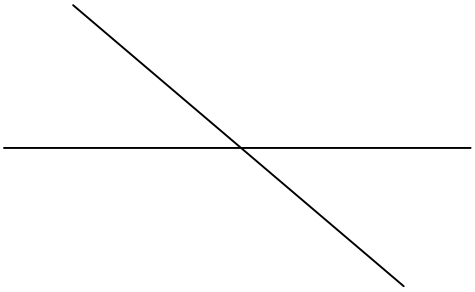
Exercise 2

Angles

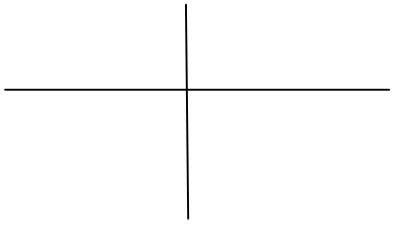
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1) Which of the following pairs of lines are perpendicular?

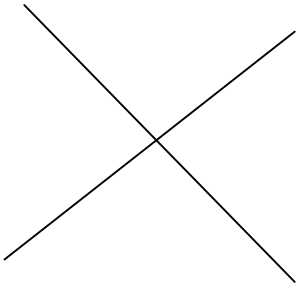
a)



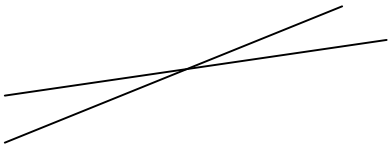
b)



c)



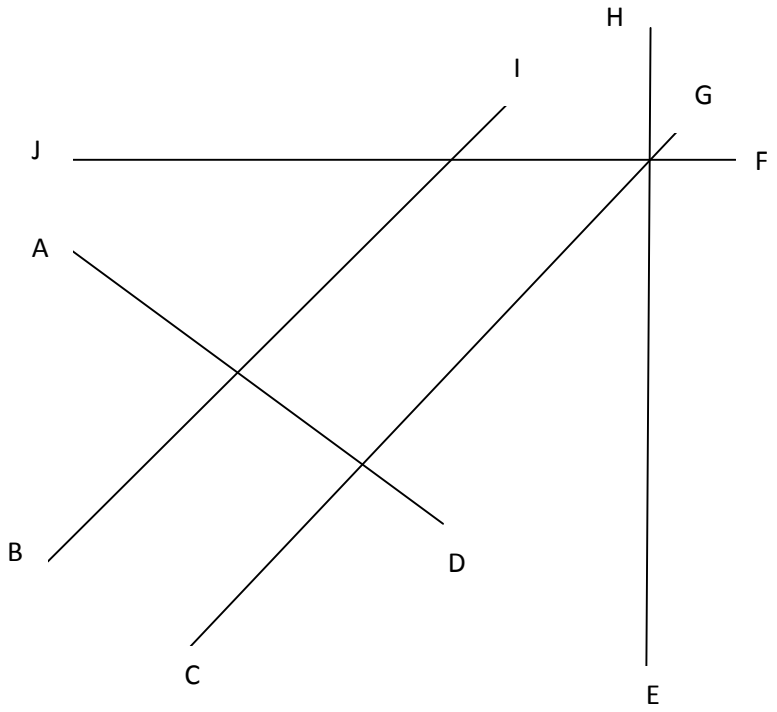
d)



B and c

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2) In the following diagram name all the perpendicular pairs of lines



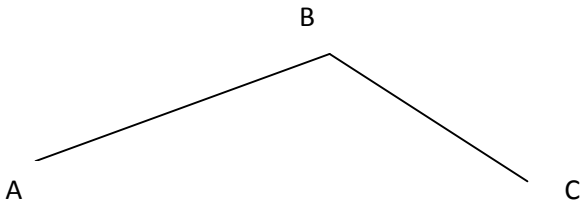
AD ⊥ BI

AD ⊥ CG

JF ⊥ EG

3) Which letter denotes the vertex in each of the following angles?

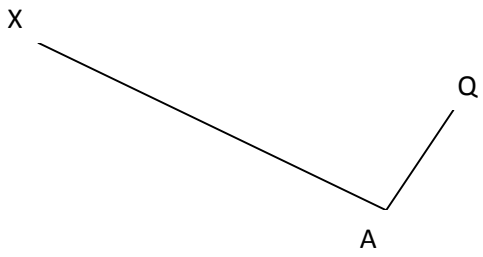
a)



B

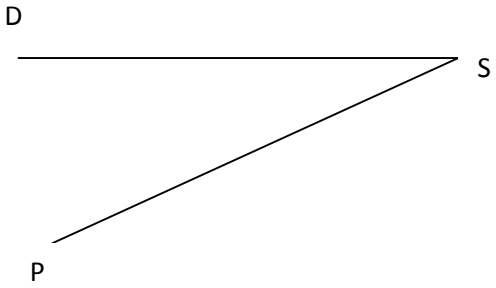
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b)



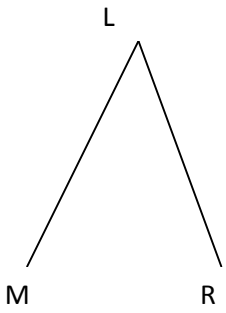
A

c)



S

d)



L

e)



T

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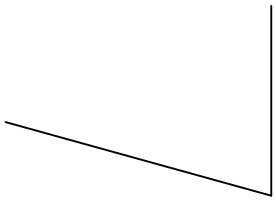
f)



X

4) Describe each of the following angles as less than right-angled, more than right angled or right-angled

a)



Less than right angled

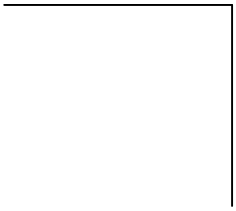
b)



Right angled

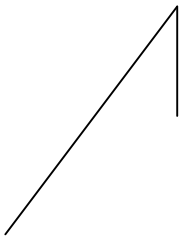
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c)



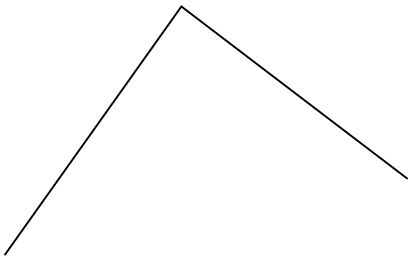
Right angled

d)



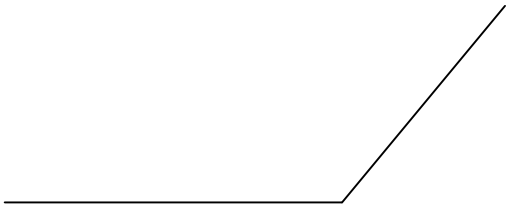
Less than right angled

e)



Right angled

f)

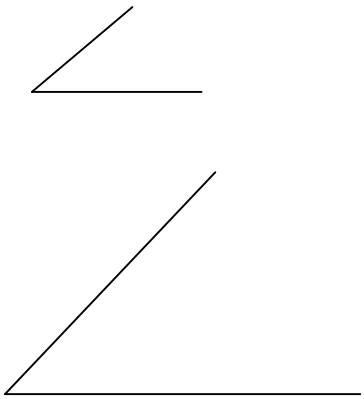


More than right angled

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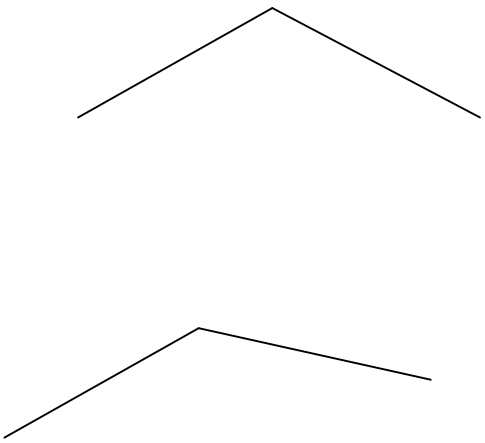
5) State whether each pair of angles are the same size

a)



Yes

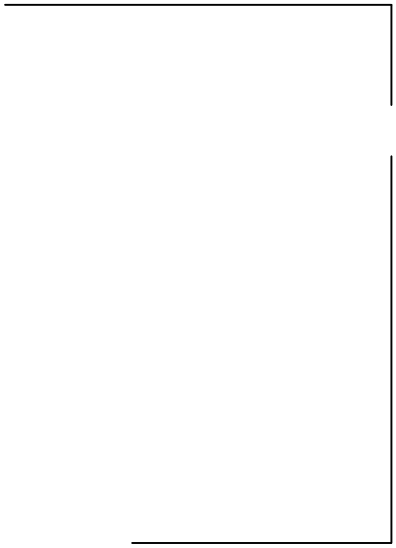
b)



No

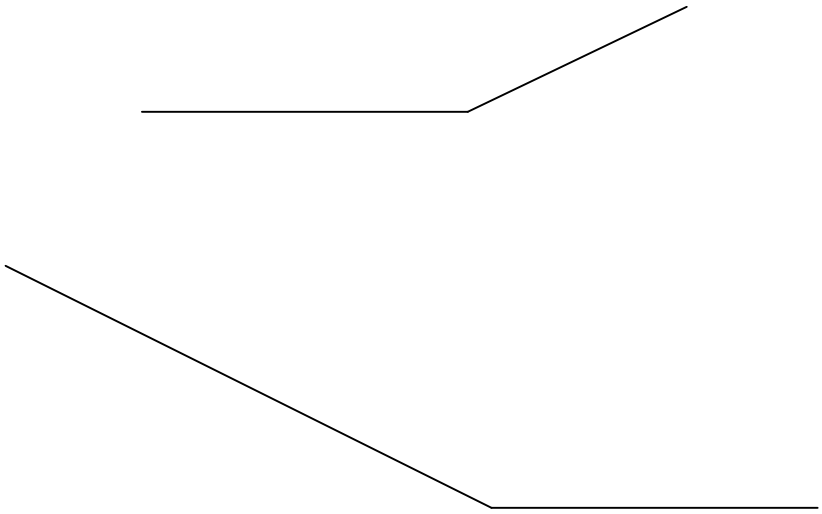
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c)



Yes

d)

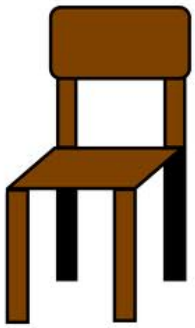


Yes

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6) Identify what parts of the following objects form angles

a)



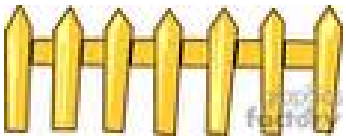
Legs to the base of the chair

Seat to the struts

Struts to the back

Back, seat, legs, struts

b)



Spikes of the fence posts

Rail to the spikes

c)



Door sides, door frame

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d)



Path

End of path to the house

Windows

Door

Roof

Chimney

e)



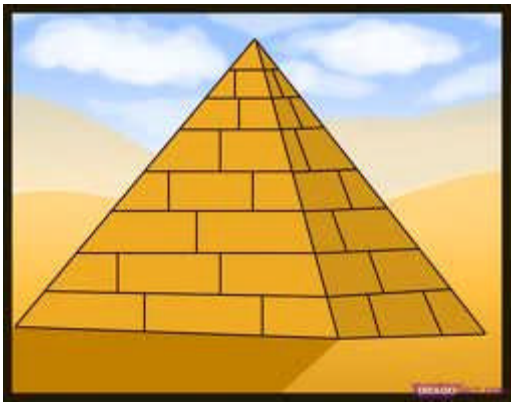
Perimeter of the sign

Letter T

White line

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f)



Base of pyramid to ground

Edges of pyramid

Faces of pyramid to each other and to the ground

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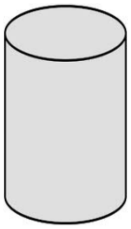
Exercise 3

2D and 3D Shapes

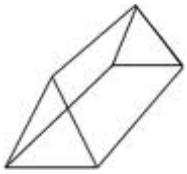
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1) Sketch the following shapes

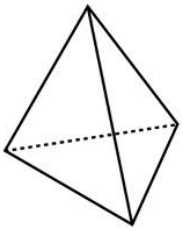
a) Cylinder



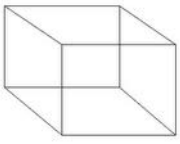
b) Triangular prism



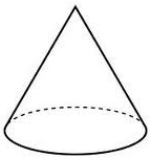
c) Triangular pyramid



d) Rectangular prism

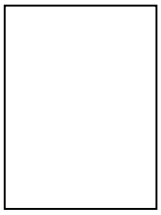


e) Cone

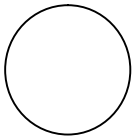


2) Sketch a cylinder from the following views

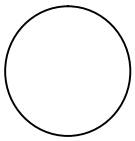
a) Side



b) Above

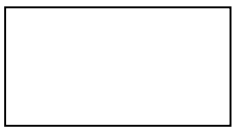


c) Below



3) Sketch a triangular prism from the following views

a) Side

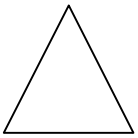


b) Below



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c) End

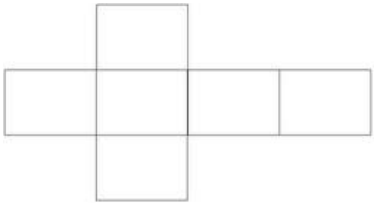


d) Above

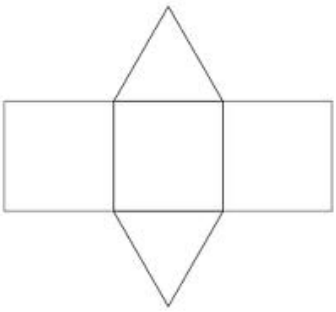


4) Draw a net of the following shapes

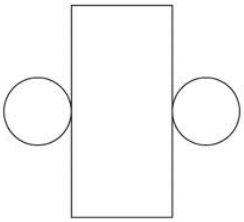
a) Rectangular prism



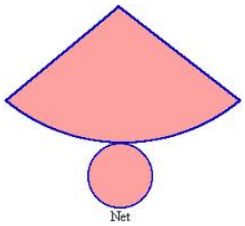
b) Triangular pyramid



c) Cylinder

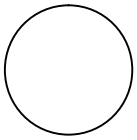


d) Cone



5) Draw and describe the shape formed when a cross section parallel to the base is taken of the following

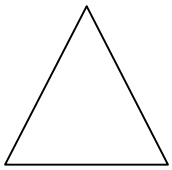
a) Cylinder



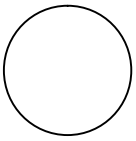
b) Rectangular prism



c) Triangular pyramid



d) Cone



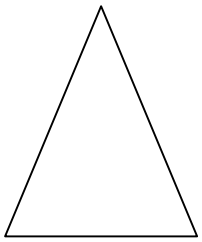
All these cross sections are the same shape as the base

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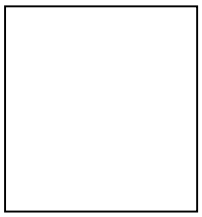
In shapes with an apex; (e.g. pyramid) the cross section is smaller than the base. In prisms the cross section is the same size as the base

6) Draw and describe the shape formed when a cross section perpendicular to the base is taken of the following

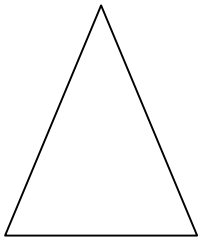
a) Cone



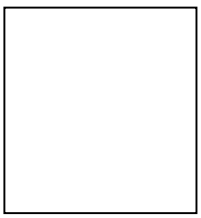
b) Triangular prism



c) Square pyramid



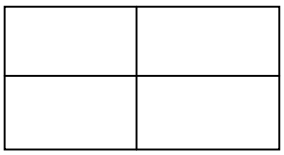
d) Cylinder



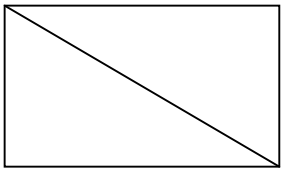
In shapes that have an apex, the cross section is a triangle. In prisms and cylinders the cross section is a rectangle

7)

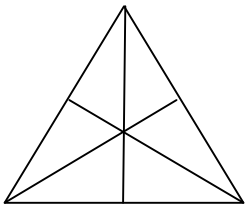
a) Draw the lines of symmetry of a rectangle



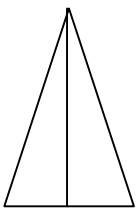
b) Draw a line through a rectangle that is not a line of symmetry



8) Draw a triangle that has all sides of equal length and draw all its lines of symmetry

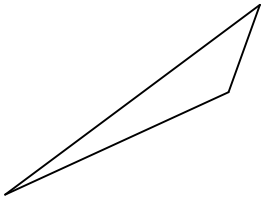


9) Draw a triangle that has 2 of its sides having equal length, and draw all its lines of symmetry



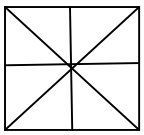
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10) Draw a triangle that has no sides of equal length and draw all its lines of symmetry

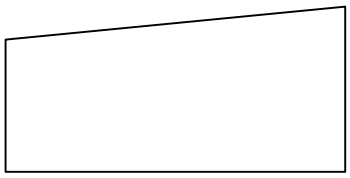


Such a triangle has no lines of symmetry

11) Draw a square and also draw all its lines of symmetry



12) Draw a four sided shape that has no sides of equal length and draw all its lines of symmetry



Any irregular shape has no lines of symmetry

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Year 4 Mathematics

Measurement

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Exercise 1

Time

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1) Write the following times in words

a)



Four twelve

b)



One thirty nine

c)



Nine thirty

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d)



Eight twenty four

2) Write the following times in two different ways. (For example seven forty-five, quarter to 8)

a)



Twelve forty five, quarter to one

b)



Ten forty, twenty to eleven

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c)



Eight fifteen, quarter past eight

d)



Six thirty, half past six

3) Convert the following to minutes

a) 1 hour

60 minutes

b) 2 hours

120 minutes

c) 1 and a half hours

90 minutes

d) Ten hours

600 minutes

e) 2 hours and fifteen minutes

135 minutes

f) 4 hours and ten minutes

250 minutes

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4) Convert the following to seconds

a) One minute

60 seconds

b) Two minutes

120 seconds

c) Five minutes

300 seconds

d) Two and a half minutes

150 seconds

e) Six minutes and 20 seconds

380 seconds

f) 1 hour

3600 seconds

5) Write each of these times as they would appear on a digital clock

a) Eight thirty

8:30

b) Six forty five

6:45

c) Quarter past three

3:15

d) Half past nine

9:30

e) Ten minutes to one

12:50

f) Quarter to 8

7:45

g) Noon

12:00

6) The main movie at the theatre shows every 2 and a half hours. If it started at seven thirty, when would the next showing begin?

10 0'clock

7) A bus goes from the city to John's street every fifteen minutes. If the last bus for the night leaves at nine o'clock, when did the second last bus leave

Fifteen minutes earlier, which is 8:45

8) A magazine is published every 2 weeks. If t was published on May

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1st, when is the next time it would be published?

May 15th

9) The American Civil War started in 1860 and went until 1865. How long did it last for?

1865 --1860 = 5 years

10) It took Alan one and a half years to sail around the world. If he left on January 1st 2010, when did he return?

July 1st 2011

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Exercise 2

Mass

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1) Convert the following to grams

a) Half a kilogram

$$\frac{1}{2} \times 1\ kg = 0.5kg = 500g$$

b) One quarter of a kilogram

$$\frac{1}{2} \times 1kg = 0.25kg = 250g$$

c) One fifth of a kilogram

$$\frac{1}{5} \times 1kg = 0.2kg = 200g$$

d) Three quarters of a kilogram

$$\frac{3}{4} \times 1kg = 0.75kg = 750g$$

e) One third of a kilogram

$$\frac{1}{3} \times 1kg = 0.33kg = 333.33g$$

2) Convert the following to kilograms

a) 500 grams

$$\frac{500}{1000} = 0.5kg$$

b) 750 grams

$$\frac{750}{1000} = 0.75kg$$

c) 250 grams

$$\frac{250}{1000} = 0.25kg$$

d) 100 grams

$$\frac{100}{1000} = 0.1kg$$

e) 1500 grams

$$\frac{1500}{1000} = 1.5kg$$

f) 1250 grams

$$\frac{1250}{1000} = 1.25kg$$

g) 3500 grams

$$\frac{3500}{1000} = 3.5kg$$

3) Add the following giving your answer in kg

a) 500g + 500g

$$= 1000g = 1kg$$

b) 700g + 700g + 600g

$$= 2000g = 2kg$$

c) 200g + 800g

$$= 1000g = 1kg$$

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d) One and a half kg plus half a kg

$= 1.5kg + 0.5kg = 2kg$

e) 750g + 750g

$= 1500g = 1.5kg$

f) One and a half kg plus one and a half kg

$= 1.5kg + 1.5kg = 3kg$

4) Write the following in kg

a) Four lots of 500g

$4 \times 500g = 2000g = 2kg$

b) Three lots of 500g

$3 \times 500g = 1500g = 1.5kg$

c) Half of 4kg

$\frac{1}{2} x 4kg = 2kg$

d) Five and a half kg subtract two and a half kg

$5.5kg - 2.5kg = 3kg$

e) One half of 5kg

$\frac{1}{2} \times 5kg = 2.5kg$

5) Eric has a bag of marbles. Each marble weighs 200g and he has 10 of them. If John's marbles each weigh 400g, how many does he need to have the same weight of marbles as Eric?

Eric has $10 \times 200g = 2000g = 2kg$ of marbles

$5 \times 400g = 2000g$

Therefore John needs five 400g marbles

6) Four men each carry a bag of rocks weighing 250g. How many kg do they carry between them?

$4 \times 250g = 1000g = 1kg$

7) John has $5 and wants to buy as much paper as he can. Each 100g of paper costs 50 cents. How much paper can he buy?

John has 10 lots of 50 cents ($5), so he can buy 10 lots of 100g

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$10 \times 100g = 1000g = 1kg$

John can buy 1kg of paper

8) Three books weigh 250g, 300g and 600g. How much do the books weigh together?

$250g + 300g + 600g = 1150g = 1.15kg$

9) Peter has three weights: two of them weigh 400g and the other weighs 700g. Alan has two weights: one weighs 1kg and the other 500g. Who has more weight?

Peter's total of weights is $400g + 400g + 700g = 1500g = 1.5kg$

Alan's total of weights is $1kg + 0.5kg = 1.5kg$

Peter and Alan have the same weight

10) Thomas eats 500g of a 750 g steak, while his Dad leaves 100g of his. How much steak is left in total?

Thomas has $750g - 500g = 250g$ of his steak left

$250g + 100g = 350g$ of steak left in total

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Exercise 3

Length, Perimeter & Area

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1) Convert the following to metres (e.g. 1m 50cm = 1.5m)

a) 1 m 25cm

$$25cm = \frac{25}{100}m = 0.25m$$

$$1m + 0.25m = 1.25m$$

b) ½ m

$$\frac{1}{2} \times 1m = 0.5m$$

c) 2 m 50cm

$$50cm = \frac{50}{100}m = 0.5m$$

$$2m + 0.5m = 2.5m$$

d) 3m 60cm

$$60cm = \frac{60}{100}m = 0.6m$$

$$3m + 0.6m = 3.6m$$

e) 2m 75cm

$$75cm = \frac{75}{100}m = 0.75m$$

$$2m + 0.75m = 2.75,$$

f) 80cm

$$80cm = \frac{80}{100}m = 0.8m$$

2) Convert the following to m and cm (e.g. 1.5m = 1m 50cm)

a) 1.25m

$$1.25m = 1m + 0.25m$$

$$0.25m \times 100 = 25cm$$

$$1.25m = 1m\ 25cm$$

b) 600 cm

$$600cm = \frac{600}{100}m = 6m$$

c) 2.75m

$$2.75m = 2m + 0.75m$$

$$0.75m = 0.75 \times 100cm$$
$$= 75cm$$

$$2.75m = 2m\ 75cm$$

d) 0.5m

$$0.5m = 0.5 \times 100cm$$
$$= 50cm$$

e) 4.2m

$$4.2m = 4m + 0.2m$$

$$0.2m = 0.2 \times 100cm$$
$$= 20cm$$

$$4.2m = 4m\ 20cm$$

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f) 1.05m

$1.05m = 1m + 0.05m$

$0.05m = 0.05 \times 100cm$
$= 5cm$

$1.05m = 1m\ 5cm$

3) Graham is 1.6m tall, while his dad is 2 metres. How much taller is Graham's dad in metres?

$2m - 1.6m = 0.4m$

4) A square has side length of 1 metre, what is its area?

$1m \times 1m = 1m^2$

5) Would the area of the following be approximately equal to 1 square metre, less than 1 square metre, or more than 1 square metre?

- The floor of a kitchen

  More

- A window

  About equal

- A stamp

  Less

- A coffee table

  About equal

- A lawn

  More

- A field

  More

- A car door

  About equal

6) Describe how to calculate the perimeter of a shape

Measure the distance around the outside of the shape

7) Calculate the perimeter of each of the following rectangles

a) Side lengths 1m and 2m

$1m + 2m + 1m + 2m$
$= 6m$

b) Side lengths 2m and 3m

$2m + 3m + 2m + 3m$
$= 10m$

c) Side lengths 5m and 4m

$5m + 4m + 5m + 4m$
$= 18m$

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d) Side lengths 1.5m and 2m

$1.5m + 2m + 1.5m + 2m$
$= 7m$

e) Side lengths 1m 50cm and 2m

$1m\ 50cm = 1.5m$

Rectangle and hence answer are same as previous question

f) Side lengths 50cm and 1m

$50cm = 0.5m$

$0.5m + 1m + 0.5m + 1m$
$= 3m$

8) Calculate the area of each of the following rectangles

a) Side lengths 1m and 2m

$1m \times 2m = 2m^2$

b) Side lengths 2m and 3m

$2m \times 3m = 6m^2$

c) Side lengths 5m and 4m

$5m \times 4m = 20m^2$

d) Side lengths 1.5m and 2m

$1.5m \times 2m = 3m^2$

e) Side lengths 1m 50cm and 2m

$1m\ 50cm = 1.5m$

Rectangle and hence area is same as previous question

f) Side lengths 50cm and 1m

$50cm = 0.5m$

$0.5m \times 1m = 0.5m^2$

9) There are two pieces of wood on the ground. One has a length of 1m and a width of 4m, the other is a square piece of side length 2m. Which piece of wood has a bigger area? Which piece of wood has the bigger perimeter?

Area of first piece =

$1m \times 4m = 4m^2$

Area of second piece =

$2m \times 2m = 4m^2$

The two pieces have the same area

Perimeter of first piece =

$1m + 4m + 1m + 4m = 10m$

Perimeter of second piece =

$2m + 2m + 2m + 2m = 8m$

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First piece has larger perimeter

10) A man walked around a lounge room that was 3m long and 2m wide. How far did he walk?

Perimeter =

$3m + 2m + 3m + 2m = 10m$

11) The man from question 10 wishes to carpet his lounge room. How many square metres of carpet will he need?

Area = $3m \times 2m = 6m^2$

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Exercise 4

Volume & Capacity

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1) Estimate the capacity in litres of each of the following?

NOTE the following are estimates only

- A milk carton

  Usually 1 litre

- A car's petrol tank

  Anywhere from 50 to 100 litres

- A bath

  Around 200 litres

- A large bottle of soft drink

  2 litres

- A swimming pool

  Depends on type of pool: a backyard pool could be around 250,000 litres to an Olympic pool that has a capacity of around 5 million litres

- A kitchen sink

  Around 20 litres

2) Convert the following to mL

a) 1.25 L

$$1.25l = 1.25 \times 1000mL$$
$$= 1250ml$$

b) 2.6L

$$2.6l = 2.6 \times 1000ml$$
$$= 2600ml$$

c) 0.75L

$$0.75l = 0.75 \times 1000ml$$
$$= 750ml$$

d) 3.9L

$$3.9l = 3.9 \times 1000ml$$
$$= 3900ml$$

e) 2.24L

$$2.24l = 2.24 \times 1000ml$$
$$= 2240ml$$

f) 8L

$$8l = 8 \times 1000ml$$
$$= 8000ml$$

3) Convert the following to Litres

a) 4000mL

$$4000m = (4000 \div 1000)l$$
$$= 4l$$

b) 2500mL

$$2500m = (2500 \div 1000)l$$
$$= 2.5l$$

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c) 1250mL

$$1250m = (1250 \div 1000)l$$
$$= 1.25l$$

d) 4750mL

$$4750m = (4750 \div 1000)l$$
$$= 4.75l$$

e) 10000mL

$$10000ml = (10000 \div 1000)l$$
$$= 10l$$

4) How much liquid is wasted if 500mL is added to a 1 litre container that already contains 750mL?

$750ml + 500ml = 1250ml$

The container overflows by $250ml$

5) To fill a 2L container, how much liquid needs to be added if it currently contains 1.4 litres?

$2l - 1.4l = 0.6l = 600ml$

$600ml$ should be added

6) Bill poured 600mL of water into a bowl, Tom poured a further 500mL and Peter poured 900mL. How much water was in the container?

$$600ml + 500ml + 900ml$$
$$= 2000ml = 2l$$

7) A 1 litre container is filled to the top with water. One hundred 1cm^3 blocks are thrown into the container and water overflows as a result of this. How much water is left in the container?

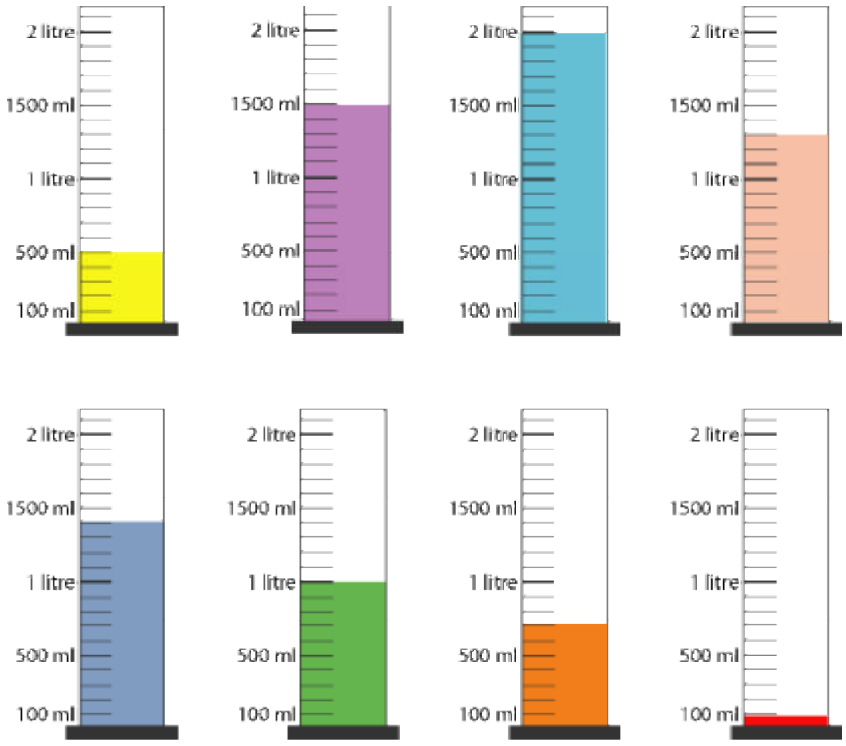
$100 \times 1cm^3 = 100cm^3$

$100cm^3 = 100ml$

Therefore there is $900ml$ of water left in the container

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8) How much liquid is in the following cylinders?



$500ml$ $1500ml = 1.5l$ $2l$ $1300ml = 1.3l$

$1400ml = 1.4l$ $1l$ $700ml = 0.7l$ $100ml = 0.1l$

9) Stacks of 1 cm blocks are built. How much water would they displace from a container if they were dropped in?

(Each block is $1cm^3$)

a) 2 rows and 3 columns

$2 \times 3 = 6\ blocks = 6cm^3 = 6ml$

b) 4 rows and 5 columns

$4 \times 5 = 20\ blocks = 20cm^3 = 20ml$

c) 6 rows and 3 columns

$6 \times 3 = 18\ blocks = 18cm^3 = 18ml$

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d) 3 rows and 6 columns

$3 \times 6 = 18\ blocks = 18cm^3 = 18ml$

e) 10 rows and 10 columns

$10 \times 10 = 100\ blocks = 100cm^3 = 100ml$

f) 30 rows and 30 columns

$30 \times 30 = 900\ blocks = 900cm^3 = 900ml$

10) In a fridge there were five 250 mL cans of soft drink. How much soft drink was there altogether?

$5 \times 250ml = 1250ml = 1.25l$

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